

Clostridium difficile in foods and animals: history and measures to reduce exposure.

.

Many articles have summarized the changing epidemiology of *Clostridium difficile* infections (CDI) in humans, but the emerging presence of *C. difficile* in foods and animals and possible measures to reduce human exposure to this important pathogen have been infrequently addressed. CDIs have traditionally been assumed to be restricted to health-care settings. However, recent molecular studies indicate that this is no longer the case; animals and foods might be involved in the changing epidemiology of CDIs in humans; and genome sequencing is disproving person-to-person transmission in hospitals. Although zoonotic and foodborne transmission have not been confirmed, it is evident that susceptible people can be inadvertently exposed to *C. difficile* from foods, animals, or their environment. Strains of epidemic clones present in humans are common in companion and food animals, raw meats, poultry products, vegetables, and ready-to-eat foods, including salads. In order to develop science-based prevention strategies, it is critical to understand how *C. difficile* reaches foods and humans. This review contextualizes the current understanding of CDIs in humans, animals, and foods. Based on available information, we propose a list of educational measures that could reduce the exposure of susceptible people to *C. difficile*. Enhanced educational efforts and behavior change targeting medical and non-medical personnel are needed.